

Session 8 (AFRM):

Volatility based VaR models: In these models, we assume the distribution of P/Ls as Normal distribution or t-distribution, etc.

$$r_t \sim N(0, \sigma^2)$$

mean = 0
variance = σ^2 (2nd moment)
 σ^2 is taken out as square root of σ^2 , i.e., $\sqrt{\sigma^2} = \sigma$

$$r_t \sim \sigma \cdot N(0, 1)$$

Assume, $Z = N(0, 1)$

$$\rightarrow r_t = \sigma \cdot Z$$

$\downarrow$
P/L

$\therefore$ For long position,

$$\begin{aligned} 95\% \text{ 1-day } \text{VaR}_{t+1} &= \text{quantile}(r_t, 5\%) \\ &= \text{quantile}(\sigma \cdot Z, 5\%) \\ &= \sigma * \text{quantile}(Z, 5\%) \end{aligned}$$

$$95\% \text{ 1-day } \text{VaR}_{t+1} = -1.645 * \sigma_{t+1} \quad (\% \text{-VaR})$$

$\downarrow$
forecasted volatility

$$95\% \text{ 1-day } \text{VaR}_{t+1} = -1.645 * \sigma_{t+1} * V_{P,t} \quad (\text{Monetary VaR})$$

$\downarrow$
value of position at time t.

$$99\% \text{ 1-day } \text{VaR}_{t+1} = -2.33 * \sigma_{t+1} * V_{P,t}$$

$$99.5\% \text{ 1-day } \text{VaR}_{t+1} = -2.58 * \sigma_{t+1} * V_{P,t}$$

For short position,

$$95\% \text{ 1-day } \text{VAR}_{t+1} = -1.645 * \sigma_{t+1} * V_{P,t}$$

$$99\% \text{ 1-day } \text{VAR}_{t+1} = -2.33 * \sigma_{t+1} * V_{P,t}$$

$$99.5\% \text{ 1-day } \text{VAR}_{t+1} = -2.58 * \sigma_{t+1} * V_{P,t}$$

① Simple variance based approach: In this approach, the forecasted variance is estimated as sample variance based on historical returns.

$$\sigma_{t+1}^2 = \frac{\sum_{i=1}^t (R_i - \bar{R})^2}{t-1}$$

→ then generate forecasted volatility as:-

or, we can take $\sigma_{t+1} = \sqrt{\sigma_{t+1}^2}$ sample std of historical returns to get estimate of forecasted volatility.

$$\therefore 95\% \text{ 1-day } \text{VAR}_{t+1} = -1.645 * \sigma_{t+1} * V_{P,t}$$

$$99\% \text{ 1-day } \text{VAR}_{t+1} = -2.33 * \sigma_{t+1} * V_{P,t}$$

$$99.5\% \text{ 1-day } \text{VAR}_{t+1} = -2.58 * \sigma_{t+1} * V_{P,t}$$

Backtesting: Using moving window of 250 days of historical returns, we will generate forecast of volatility by estimating sample std & then estimating VAR & compare it with actual P/L.

Drawback of simple variance based approach:

When we estimate variance using simple variance based approach, we take an average of last many days (e.g. 250 days) of variance. Hence, the

sudden change in volatility is ~~not~~ not appropriately captured & is captured in lagged fashion. because we give equal weight to all the previous information. The right way will be to give high weight to current information & lower weight to previous information.

J P Morgan proposed "RiskMetrics" model based on "Exponential Weighted Moving Average (EWMA)" model. In this model high weight is given to recent information & weight decline exponentially for historical information while estimating variance.

② RiskMetrics (RM) model: It is based on EWMA model. According to EWMA model, the variance is given as:-

$$\sigma_{t+1}^2 = (1-\lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-i}^2$$

weight given to different days variance

where, $\lambda = 0.94$

$$\sigma_{t+1}^2 = (1-\lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-i}^2$$

$$\sigma_{t+1}^2 = (1-\lambda) \cdot \lambda^0 \cdot R_{t-0}^2 + (1-\lambda) \sum_{i=1}^{\infty} \lambda^i R_{t-i}^2$$

$$\sigma_{t+1}^2 = (1-\lambda) \cdot R_t^2 + \lambda (1-\lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-1-i}^2$$

$$(1-\lambda) \sum_{i=1}^{\infty} \lambda^i R_{t-i}^2$$

$$\lambda (1-\lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-1-i}^2$$

First value: $i=1$

$$(1-\lambda) \lambda^1 R_{t-1}^2 \\ = (1-\lambda) \lambda \cdot R_{t-1}^2$$

First value: $i=0$

$$\lambda (1-\lambda) \cdot \lambda^0 \cdot R_{t-1-0}^2 \\ = \lambda (1-\lambda) \cdot R_{t-1}^2$$

both are same.

Second value: $i=2$

$$(1-\lambda)\lambda^2 \cdot R_{t-2}^2$$

Second value: $i=1$

$$\lambda(1-\lambda)\lambda^1 \cdot R_{t-1-1}^2$$

$$= (1-\lambda)\lambda^2 \cdot R_{t-2}^2$$

both are same

both sequences will have same sequence

$$\sigma_{t+1}^2 = (1-\lambda)R_t^2 + \lambda \left[(1-\lambda) \sum_{i=0}^{\infty} \lambda^i R_{t-1-i}^2 \right] = \sigma_t^2$$

$$\sigma_{t+1}^2 = (1-\lambda)R_t^2 + \lambda \sigma_t^2$$

$$= (1-0.94)R_t^2 + 0.94\sigma_t^2$$

$$\sigma_{t+1}^2 = 0.06R_t^2 + 0.94\sigma_t^2$$

Estimating VaR using RiskMetrics:

→ First generate forecast of variance,

$$\sigma_{t+1}^2 = (1-\lambda)R_t^2 + \lambda \sigma_t^2$$

$$= 0.06R_t^2 + 0.94\sigma_t^2$$

→ Estimate forecasted volatility,

$$\sigma_{t+1} = \sqrt{\sigma_{t+1}^2}$$

→ Estimate VaR,

$$95\% \text{ 1-day } VaR_{t+1} = -1.645 * \sigma_{t+1} * \sqrt{P_{t+1}}$$

$$99\% \text{ 1-day } VaR_{t+1} = -2.33 * \sigma_{t+1} * \sqrt{P_{t+1}}$$

$$99.5\% \text{ 1-day } VaR_{t+1} = -2.58 * \sigma_{t+1} * \sqrt{P_{t+1}}$$

Note: For 1st day, we take Return^2 (squared return) as proxy for 1st day variance. From 2nd day onward, we use the RiskMetrics formula.

